

Policy Q&A Assistant — Solution Overview

Problem

People struggle to understand and follow a wide range of policies — not just **corporate rules**, but also **insurance policies**, **school policies**, **government scheme guidelines**, **bank terms**, and **travel rules**. Policies are often:

- Long and complex
- Scattered across PDFs and intranet pages
- Hard to search
- Updated without clear communication

Individuals spend hours Googling, calling support, or asking others to interpret complex policy language. This leads to:

- Reduced productivity
- Policy misunderstanding
- Compliance risks
- Increased load on support teams

Solution

Policy Q&A Assistant — An AI-powered assistant that allows employees to upload or reference company policies and ask natural language questions.

The system:

- Extracts and indexes policy documents

- Answers questions grounded strictly in policy text
- Shows page-specific citations
- Provides clear, compliant, factual responses

What it Does

Feature	Benefit
Upload policy PDF	Centralizes knowledge
AI answers questions	Employees get clarity instantly
RAG + page citations	Trustworthy, document-backed answers
"Not legal advice" guardrail	Safe, compliant usage
Document updates supported	Always current

Workflow

1. **Upload Policy PDF**
2. **Extract text with page numbers**
3. **Chunk text into policy sections**
4. **Generate embeddings** (vector representations)
5. **Store embeddings in vector database**
6. **User asks question**
7. **Retrieve top relevant policy excerpts (RAG)**
8. **Build contextual prompt with citations**
9. **LLM generates grounded answer**
10. **Response returned with page references & disclaimer**

Architecture

- Frontend UI: Streamlit
- Backend: Python
- PDF Parsing: PyMuPDF
- Embeddings Model: Sentence Transformers (MiniLM)
- Vector Store: FAISS
- LLM: OpenRouter API (Llama/Claude)
- Environment: `.env` for secure API keys

Tech Stack

Layer	Tool
App Framework	Streamlit
Language	Python
PDF Extraction	PyMuPDF
OCR (optional future)	Tesseract
Embeddings	Sentence Transformers
Vector Search	FAISS
LLM	OpenRouter (Llama 3 / Claude)
Secrets Management	dotenv
Deployment	Local / Cloud

Safety & Compliance

- Does not hallucinate policy content
- Always cites the policy page
- Returns "Not specified" when needed
- Adds "Not legal advice" disclaimer

Why this Solves the Problem

Challenge	Solution
Employees can't find policy answers quickly	Instant policy lookup
HR/IT overloaded with repeated questions	Reduced support burden
Risk of misinterpretation	Verified, citation-backed answers
Policies constantly updated	Re-indexable, version-safe
Scattered PDFs, wiki docs	Centralized Q&A interface

Business Impact

- 🚀 Faster onboarding & policy understanding
- 🗝️ Stronger compliance culture
- ⌚ Saves internal productivity hours
- 🤝 Better employee support & autonomy

This AI-powered assistant transforms static policy documents into an interactive knowledge system, providing accurate, citation-backed answers instantly — improving compliance, reducing HR load, and empowering employees.

"No more digging through PDFs — just ask and get the exact page citation."